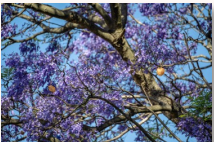
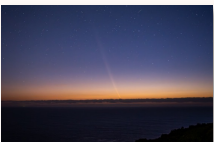
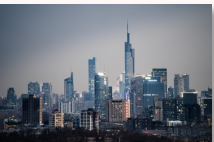
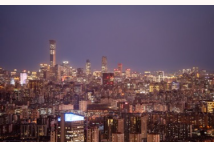


Support Set of Target User

	<i>user score</i> y	<i>global prior</i> $g(x)$	$r = y - g(x)$	
	3.6	4.2	- 0.6	→
	4.8	4.5	+ 0.3	→
	3.8	3.0	+ 0.5	→
	3.8	2.6	+ 1.2	→
	3.2	4.4	- 1.2	→
	3.7	3.8	- 0.1	→
	4.7	4.3	+ 0.4	→

Residual Scoring and Score-bin Pooling

1

Compute Residual r

$$r = y_{user} - g(x)$$

2

Pooled by Residual Sign/Score Bin

Group residuals by sign(positive / negative) and by score bins, then pool to form memory tokens.

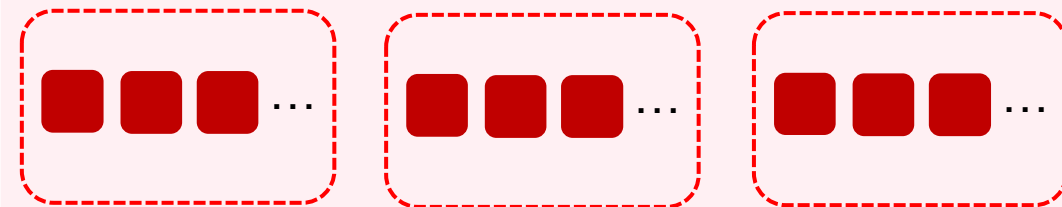
Positive Residual Memory M_u^+

liked more than global prior

low positive residual

mid positive residual

high positive residual



positive pooled token

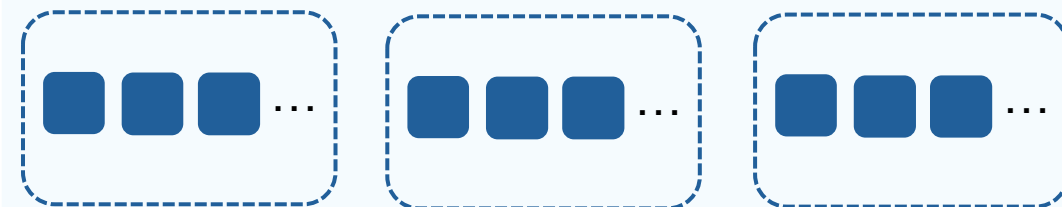
Negative Residual Memory M_u^-

liked less than global prior

low negative residual

mid negative residual

high negative residual



negative pooled token